
Data Structures

BS (CS) _Fall_2024

Lab_03 Tasks



Learning Objectives:

1. Multi-Dimensional Array
2. Multi-Dimensional Array with Templates

Lab Tasks

Submission Instructions

1. Create a separate **.h** and **.cpp** file for each class.
2. Now create a new folder with name *ROLLNO_SEC_LAB01* e.g. **i23XXXX_A_LAB03**(Also name your **cpp** file like this.)
3. Move all of your files to this newly created directory and compress it into a **.zip file**. Return the test file as well.
4. Now you have to submit this zipped file on Google Classroom.
5. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.
6. Plagiarism or AI generated code in the Lab Task will result in **zero** marks in the whole category.

Q 1. Write a C++ program using a template to rotate a 'm x n' matrix by 90 degrees clockwise in-place.

For Example the given matrix:

Array = [3 4 2 5 6 1 7 8 9]

After the 90 degree rotation

Array = [7 5 3 8 6 4 9 1 2]

Prototype: `T** rotateMatrix90DegreesClockwise(T (&matrix)[SIZE][SIZE])`

Q2. Implement a C++ program using templates to analyze the performance of cricket players based on their recent match statistics. Use the data to select the top 10 players for the next match based on their performance.

Dataset Description:

You have performance data for 12 cricket players over their recent 10 matches. Each player's performance is recorded as runs scored in each match. You will store this data in a 2D array where each row corresponds to a player, and each column corresponds to the runs scored in a specific match.

Tasks:

1. **Input Data:**

- Create a 2D array to store the runs scored by 12 players over 10 matches.
- Prompt the user to input the runs scored by each player in each match.

2. **Compute Statistics:**

- **Total Runs per Player:** Calculate the total runs scored by each player across all 10 matches.
- **Average Runs per Match per Player:** Calculate the average runs scored by each player per match.
- **Best 10 Players:** Select the top 10 players based on their total runs scored across all matches.

3. **Output Data:**

- Display the total runs and average runs per match for each player.
- Print the list of the top 10 players based on their total runs scored, sorted in descending order. Also mention the player number of those 10 players.

Prototype:

T* **inputPlayerData**(T runs[][numMatches], int numPlayers);

T* **computeStatistics**(const T runs[][numMatches], int numPlayers, T totalRuns[], T avgRuns[]);

int* **sortPlayersByTotalRuns**(T totalRuns[], int playerIndices[], int numPlayers);

T **printTop10Players**(const T totalRuns[], const int playerIndices[], int numPlayers);

Q3. Write a C++ program to calculate the determinant of a 2D square matrix (m x n). You must implement a function template that takes a square matrix (2D array) as input and returns its determinant.

Prototype: T** calculateDeterminant(T matrix[SIZE_M][SIZE_N])

Q4. Write a C++ program using a template to traverse a 2D matrix in a spiral order and return the result as a single array. You should handle matrices of arbitrary sizes.

Example:

Given the matrix:

Array = [1 4 7 10 2 3 5 6 8 11 9 12]

The output should be:

Result = [1 4 7 10 11 12 9 6 3 2 5 8]

Prototype: T* spiralOrder(const T (&matrix)[rows][cols], int& size)